

Monkey Pox Detection using Deep Learning

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Abstract—In this research article, the problem of detecting and identifying a new disease, referred to as "Monkey Pox," is discussed. The widespread outbreak of the newly discovered disease that has been introduced has caused thousands of individuals to become ill and has even been linked to deaths. The effects of the disease might continue for up to three or four weeks in a victim. In the present circumstances, there are no such methods that can analyse the sickness. In this research, we are going to work on the detection of this disease by employing a number of different Machine Learning models, such as Monkey Pox Skin Lesion Datasets (MSLD), KNN Analysis, Ada Booster Classifier, Naive Bayes Classifier, and Decision Tree.

Keywords—Monkey Pox, Monkey Pox Skin Lesion Datasets, Machine Learning, Decision Tree.

I. INTRODUCTION

Monkey Pox is the recent disease that has been outbreak in the world. Monkey Pox is a rare disease that is mainly caused by the Monkey Pox virus. It is also part of the virus family called varola virus, the kind of viruses where there could be risk of getting small pox. As the disease was named as Monkey Pox the source of disease is unknown. It was first outbreak in month of May this year and the first case has been found in Africa Continent. From mid of the may it has been reportedly increased in more than 40 countries worldwide too. African rodents and some clinical researches has shown that this is more challenging since it has some similarities with chickenpox and measles. In fact, it has much world of difference comparing with other diseases. People with Monkey Pox, faces lots of rashes, this rashes are going to increase in several stages, even including the scabs before healing. These rashes must look like simple and can be become most complex, itchy and painful as increases. Recently a 22year old Man from Kerala, India has died because of this disease because of lack of necessary precautions. The Monkey Pox can be caused by those of animals and infected people whoever has the disease. Such as the, by sharing or roaming the things with the effected people. It takes around one to three weeks to spread from effected

people after exploring from them. People with less and improper immune system can effected and high chances of getting Monkey Pox. If anyone in their family has such cases has to concern immediately to the doctor.

Being hygienic can also useful to prevent the occurring the cause of Monkey Pox. If any person occurred by the Monkey Pox need to be self-quarantined for couple of days. Monkey Pox is also has high impact to affect not only on humans but also in animals, mammals, plants as well. The doctors has to prescribe particular medicines, vaccines and some necessary precautions to get rid of Monkey Pox. But the Monkey Pox is the rare viral disease of people as it occurs mostly of the skin diseases only. While sometimes it becomes too hard for everyone by travelling from places to places and countries to countries or by becoming more social to everyone. Once the monkey pox has occurred to anyone it will be lasts from couple of days to 3-4 weeks as it depends upon the type of rashes it has been occurred and later on there may be chance of disappearing the rashes and might be reduced. But currently there are no checkups, treatments for the cure of this outbreak Monkey Pox. Even there is high chances of death for people with Monkey Pox. But mostly in researches it has also been considered that it belongs to the small pox families. It may not be severe to the people but still it results to death. Infact some particular persons can pass the rumors about such diseases that there is scope of getting less effected by diseases etc. But we should not trust such people. As currently there is no cure for disease so, the people whoever effected or has doubt of monkey pox need to be have checkup in hospitals for early stage of detection of Monkey Pox disease as to prevent from spread.

II. MONKEY POX CASES

The following is the data collected for out breaking of monkey pox cases that has been facing worldwide currently in which majority of cases has been seen in following countries as shown in Fig 1.

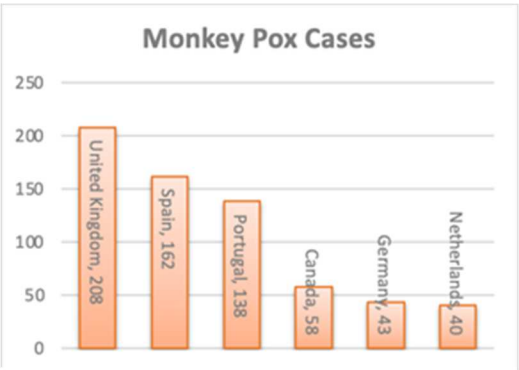


Figure 1 Worldwide number of monkey pox cases seen

It is to be found that the United Kingdom comes first with majority of cases 208 from Fig 1 followed by Spain, Portugal, Canada, Germany and the Netherlands comes last by least cases of 40.

As shown in figures , the effect of Monkey Pox on Human especially on Human skin is shown as above Fig (a) and Fig (b) it will definitely takes some days to pass from monkey to human skin and then slowly happens like rashes to human skin and then occurs like bumps that is visible over the skin due to rashes it can be occur huge amount on the part on skin over the skin as shown in fig 2(a) or else it can be occur as the single one or two rashes bumps at the particular area of skin as shown in fig 2(b).This is all about effect of monkey pox on humans skin.



Figure 2 contains the following sub, Fig (a) Effect of Monkey Pox on Human Skin Fig(b) Effect of Monkey Pox on particular part of skin, Fig(c) Effect of Monkey Pox on group of animals Fig (d) Effect of Monkey Pox on the particular pet dog

Monkey Pox is very infectious disease that not only effects on humans but as every living being such as animal, plants etc. Monkey Pox can get on some specified animals dangerously. Infact humans too are the main responsible for spreading of Monkey Pox to animals at various cases. The other thing is the spread can be happen through animals in such a way that is it can be hugging, kissing, licking, by sharing food respiration etc. Mostly the small Mammals only

can carry the large number of disease from animal to animal and sometimes animal to human. Infact whenever they are effected even they must quarantined as well in separate place.

It's been noticed mostly that the Monkey Pox has been fastly transferring from human to dogs. As the dogs are the pet animals for the majority of people in the world. However, the first case has been found in USA. As the virus has been transferred to Human to his/her pet dog. In such case if both effected by each other the first one can be effected on their skin and second one has been effected on it's anal. So for human it is directly shown upon their skin whereas coming to the animals it impacts on anal if it is passed through humans. But most of countries only wild animals are able to carry the monkey pox viruses. But it is very strange that we are seeing such scenarios in domestic animals.

III. LITERATURE REVIEW

Several Researchers has choosen these machine learning algorithms for several kinds of diseases such as Google Net, VGG16, SVM,KNN Analysis etc.

The authors of [1] have presented a method for detecting yellow grapevine in grapes. This approach uses Otsu Threshold for segmentation, while CNN was used to automatically extract features. Six distinct neural network designs, including ResNet-101, ResNet-50, Alex Net, Squeeze Net, Google Net, and Inception V3, were implemented by the suggested system. Through deep learning, the system achieved 98.96% sensitivity and 99.40% specificity.

In [2], the authors has made to utilize the deep learning capabilities to achieve automatic plant disease detection system. The classification and prediction of the diseases has done through CNN, with the training accuracy of 99% and testing accuracy of 98% specificity through deep learning.

In [3], the authors has made the CNN algorithm for four diseases and visually and classifying potatoes to get free from such symptoms. Here the augmentation of data method has done randomly. While classifying healthy tubers visually the performance of CNN was low. Most of the CNN's, when classifying these healthy tubers visually the class of silver scurf disease is occurred as misclassification. These Silver Scurf results to heavy range of tan on the respective tuber non-infected skin which shows the visualization of symptoms, which may resemble uninfected skin. Hence the accuracy has obtained 97.1% by using deep learning.

In [4], the authors has shown the accuracy and consistency of gastric cancers to improve it by using the clinical utility of deep learning model. There is pathologists severe shortage in most of developing countries, the suspicious areas are got located by the AI assistance. Thus it has improved within limited period of time regarding quality

of diagnostic. To adapt the WSIs in huge number in training phase regarding various kinds of tumors with the pixel level accuracy. The deep level CNN algorithm is the domain specific data generated pixels for the Artificial Intelligence Model is the second most aim of this work. Hence the accuracy has obtained 87.3 by CNN model in deep learning.

In [5], the main aim of the work is to classify the malignant and benign tissues and also detect the masses in the mammogram. The DCNN was used for feature extraction method and Alexnet was used to classify the medical images. The accuracy obtained is 80.5%.

In [6], the author has proposed the 100's of patient's dataset. It has resulted the following steps such as 1)By deep features extracting and trained ResNet101 the stretching step of contrast has employed.2)By implementing the Differential Equation method for optimal solution the deep features has been employed.3)By implementing the ECSA Algorithm on deep learning features and optimal solution is only selected.4)Both optimal vectors has been fused. 5) By comparing the proposed framework results with neural network methods. SVM, KNN, Naïve Bayes, Fine tree are the implemented classifiers. The obtained Accuracy is 87.45%.

In [7], here the cases has evaluated the classification methods for various circumstances by using deep learning algorithms whichever were possible for lung CT diseases for COVID-19 from lung CT images. From the results we can get to see that sensitivity values are lesser than the specificity values in each classification method. COVID-19 negative class type are useful to learn more about COVID-19 negative images in dataset. Here algorithms like SVM, KNN, Decision Trees, Random Forest, LDA, Naïve Bayes. The accuracy obtained is 94.4%.

In [8], the authors has choosen GPU based imputation methods for missing data which is mostly for study and researchers which is considered as benefits. For every new treatment data method, every drug case, every kind of disease the clinical data has variety kind of characteristics. So the hybrid data sets by using appropriate imputation methods so it can reduce the incomplete datasets estimation.Hence, the Accuracy obtained 57.65%.

In [9], the authors has studies about the quickly detecting of the tomato leaf disease. Depending upon the HSV image

the decomposition has been performed using the thresholding technique which is a colour based in the experiment. Hence the accuracy obtained is 81%.

In [10], the authors has mostly adapted the techniques directly Mobile-Nets, Google Nets. But still they have faced challenge and then they compared 10 different diseases image and then sorted out. Hence obtained accuracy is 90.3%.

In [11], the authors has talk through prearranged techniques regarding the detection of tomato leaf disease. Regarding color, texture, leaf edges of plant the structural strategies has been scrutinized. Hence obtained accuracy is 98.49%.

In [12], the authors has worked on enhancement of CNN model for disease prediction in tomato crop. Here they have compared 9 datasets for predicting the disease. Hence obtained accuracy is 91.2%.

In [13],the authors has used the advanced techniques such as VGG-19,VGG-16,ResNet,Inception V3 they have been performed in 2 different datasets for detecting the tomato leaf disease. Hence the obtained accuracy is 99%.

In [14], the authors has used Google-Net, VGG16 has been done by employing different computer different vision systems for detection of tomato leaf disease. Hence the obtained accuracy is 99.23%.

In [15], the authors has used 6 different types of tomato plant diseases both effected and not effected for learning purpose such as supervised learning technique for disease predicting. Hence the obtained accuracy is 97.3%.

In [16], the authors has detected the tomato fruit disease with correct calculations and assessment of mass in which the novel based vision system has also been developed. The overall acceptability has been supported by overall system for the development of industry and description in systems. The precision and accuracy also been developed by these systems only for the Heart Disease Prediction. Hence the obtained accuracy is 70.26%.[21,22]

TABLE I. METHODOLOGY FOR MONKEY POX DETECTION USING DECISION TREE

Author	Disease	Methods Algorithm	Accuracy
A. Cruz	Grapevine Yellows Symptoms	CNN, ResNet-101, ResNet-50, Alex Net, Squeeze Net, GoogleNet, and Inception V3.	98.96%
Murk Chohan	Plant Disease Detection	CNN	98%
Orly Erlich	Potato Tuber Disease Detection	CNN	97.1%
Zhigang Song	Gastric Cancer	CNN	87.3%
Maha Sharkas	Breast Cancer	DCNN, Alexnet	80.5%

Muhammad Attique Khan	Stomach Abnormalities	Fine tree, Naïve Bayes, cubic KNN, weighted KNN, cubic SVM, and Ensemble Learning	87.45%
Çinare OĞUZ	Covid-19	SVM, KNN, Decision Trees, Random Forest, LDA, Naïve Bayes	94.4%
Timur Köse	Vesicour ertal Reflux and Recurrent Urinary Tract Infection	GPU	57.65%
Haridas D. Gadade	Tomato Leaf Disease	LDA, KNN, CART, NB, SVM	81%
Azeddine Elhassouny	Tomato Leaf Diseases	CNN, Mobile-Net	90.3%
Naresh K. Trivedi	Tomato Leaf Diseases	Mobile-net, VGG16, Inception V3	98.49%
Mohit Agarwal	Tomato Leaf Disease	CNN, VGG16, Mobile-net, Inception V3	91.2%
Iftikhar Ahmed	Tomato Leaf Disease	VGG16, VGG19, ResNet, Inception V3	99%
Hareem Kibriya	Tomato Leaf Disease	Transfer Learning, VGG16, GoogleNet	99.23%
H. Sabrol	Tomato Plant Disease	CNN, Genetic Algorithm, Neuro Fuzzy	97.3%
Jaesu Lee	Tomato Fruit Disease	RCNN, FCN, RPN	99.02%
Ahmad Al Ahdal	Heart Disease	SVM, Naïve-Bayes, Decision-Tree Algorithm, Logistic Regression, RandomForest	70.26%

IV. ALGORITHMS & TECHNIQUES

Machine Learning is the process that has been done in several process till now such as by making benefitted decisions in object recognition, data base systems, network security, healthcare etc.[18-20]

A. Monkey Pox Skin Lession Datasets(MSLD)

As for every disease there must be collection of datasets so here for Monkey Pox detection the data set is provided such as Monkey Pox Skin Lession Datasets. The data is mainly either collected from Images, Websites, Portals and some of real life examples that were collected till now. Here some pre-trained models also supposed to introduced such as VGG-16, ResNet-50 and Inception V3 are employed to classify the Monkey Pox at very early stage. The main motto of this algorithm is to enhance the large number of data set of diseases to generalization of models.

B. Decision Tree

This is a kind of data mining technique algorithm useful for early detection of diseases. In this the data set is broken into sub classess and more into the smaller ones. As a result by keep on dividing the smaller ones it will reveal the type of disease. This algorithm is not sensitive. So it smoothly helps and reveal the kind of disease.

C. KNN Analysis

The most famous supervised learning method is KNN analysis which is used to diagnose and specify the specific the given data that is given in it. KNN is also helpful in detecting the unknown data and even large data but the number should be unknown.

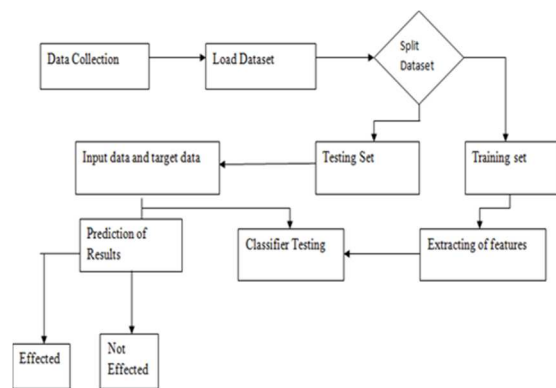


Figure 3 Step-by-step methodology for monkey pox detection

D. Ada-Booster Algorithm

In this Algorithm, the main advantage is it converts all the weak learners into the stronger ones. It changes continuously until it obtains the weight and to get precise results are obtained. It has best quality of features.

E. Naïve Bayes Classifier

The most probabilistic method that applies independent assumptions is Naïve Bayes Classifier. To detect the relationship between them the properties in this is separately considered. This classifier works fast and even with small dataset. It can give maximum accuracy of results.

F. Cluster Analysis

These techniques are always relevant regarding the disease detection. In fact the Cluster is always used to classify the disease prediction either by, whenever the data is in mixed or else it is in separated form as well. In cluster analysis, first of all it is used to select the disease of data from individuals it can also be from similar groups but it can be misunderstanding sometimes. They secondly what need to be done is measurement of the disease how depth it has gone

and lastly it will just proceeds initialization of how the things can be taken in fact it divides the original data into small to smaller one and then checks and just implements how the disease should overcome.

G. Support Vector Machine

The Support Vector Machine is used to analyse different kinds of data for disease if at all it can be from complex situations. It is most efficient regarding performing its everything in both simple and complicated data as well. It can be able to give a statistical prediction of disease that is mainly required for detection of the disease.

H. Random Forest Classifier

This classifier is used to choose arbitrary datasets, used to analyse the disease and also used to build the analysis by disease making and accordingly forecast the diseases and in fact they finalize them through the accurate results how depth the result is.

V. CONCLUSION

In this research work for detecting the monkey pox disease and predicting several deep learning techniques has implemented over that which must be noteworthy. Based on our work we have got to know that deep learning methods are always capable to determine the human diseases. Random Classifier methods are most noteworthy and performs well with large number of datasets. Regarding over fitting issues and combination of algorithms the Feature Extraction Model always performs well. We need to take necessary precautions +always regarding any disease not only monkey pox but also further diseases. If we get the doubt regarding any disease we need to concern fastly to our nearest doctors. Hence we have shown the detection of Monkey Pox Disease upto some extent. The SVM and Decision Tree Algorithm will pay a great role for the accurate prediction of disease.

REFERENCES

- [1] A. Cruz, Y. Ampatzidis, R. Pierro, A. Materazzi, A. Panattoni, L. De Bellis, et al., "Detection of grapevine yellows symptoms in *Vitis vinifera* L. with artificial intelligence", *Computers and electronics in agriculture*, vol. 157, pp. 63–76, 2019.
- [2] Murk Chohan, Adil Khan, Rozina Chohan, Saif Hassan Katpar, Muhammad Saleem Mahar et al, "Plant Disease Detection using Deep Learning", *International Journal of Recent Technology and Engineering (IJRTE)*, vol. 9, Issue 1, ISSN: 2277-3878, May 2020.
- [3] Dor Oppenheim, Guy Shani, Orly Erlich, and Leah Tsror, et al, "Using Deep Learning for Image-Based Potato Tuber Disease Detection", Published Online: 16 Apr 2019 <https://doi.org/10.1094/PHYTO-08-18-0288-R>.
- [4] Zhigang Song, Shuangmei Zou, Weixun Zhou, Yong Huang, Liwei Shao, Jing Yuan, Xiangnan Gou, Wei Jin, Zhanbo Wang, Xin Chen, Xiaohui Ding, Jinhong Liu, Chunkai Yu, Calvin Ku, Cancheng Liu, Zhuo Sun, Gang Xu, Yuefeng Wang, Xiaoqing Zhang, Dandan Wang, Shuhao Wang et al "Clinically applicable histopathological diagnosis system for gastric cancer detection using deep learning", Published: 27 August 2020, *Nature Communications* volume 11, Article number: 4294 (2020).
- [5] Dina A. Ragab, Maha Sharkas, Stephen Marshall, Jinchang Ren et al "Breast cancer detection using deep convolutional neural networks and support vector machines", Published January 28, 2019 *Electronics and Communications Engineering Department, Arab Academy for Science, Technology, and Maritime Transport (AASTMT), Alexandria, Egypt. Electronic & Electrical Engineering Department, University of Strathclyde, Glasgow, United Kingdom.*
- [6] Muhammad Attique Khan, Muhammad Shahzad Sarfraz, Majed Alhaisoni, Abdulaziz A. Albeshier, Shuihua Wang, Imran Ashraf et al "StomachNet: Optimal Deep Learning Features Fusion for Stomach Abnormalities Classification", *IEEE Volume: 8*.
- [7] Çınare OĞUZ, Mete YAĞANOĞLU et al "Determination of Covid-19 Possible Cases by Using Deep Learning Techniques", Received: 2020-07-27 15:09:24, Accepted: 2020-10-16 11:16:08, Article Type: Research Article, Volume: 25, Issue: 1, Month: February.
- [8] Timur Köse, SuÖzgür, Erdal Coşgun, Ahmet Keskinoglu and Pembe Keskinoglu et al "Effect of Missing Data Imputation on Deep Learning Prediction Performance for Vesicoureteral Reflux and Recurrent Urinary Tract Infection Clinical Study" Volume 2020 Article ID 1895076.
- [9] Haridas D. Gadade, Dr. D. K. Kirange et al "Tomato Leaf Disease Diagnosis and Severity Measurement".
- [10] Azeddine Elhassouny, Florentin Smarandache et al "Smart mobile application to recognize tomato leaf diseases using Convolution Neural Networks".
- [11] Naresh K. Trivedi, Vinay Gautam et al "Early Detection and Classification of tomato leaf disease using high performance Deep Neural Network".
- [12] Mohit Agarwal, Abhishek Singh et al "Tomato Leaf Disease Detection using Convolution Neural Network".
- [13] Iftikhar Ahmad, Muhammad Hamid et al "Optimizing Pretrained Convolutional Neural Networks for Tomato Leaf Disease Detection".
- [14] Hareem Kibriya, Rimsha Rafique et al "Tomato Leaf Disease Detection Using Convolutional Neural Network".
- [15] H. Sabrol, K. Sathish et al "Tomato Plant Disease Classification in Digital Images using Classification Tree".
- [16] Jaesu Lee, Haseeb Nazki et al "Artificial Intelligence Approach for Tomato Detection and Mass Estimation in Precision-Agriculture".
- [17] Ahmed Al Ahdal, Dr. Deepak Prashar et al "Machine Learning Based Heart Patient Scanning, Visualization and Monitoring".
- [18] T. Fadhael, P. C. H. A. Al Ahdal, M. Rakhra and D. Singh, "Design and development an Agriculture robot for Seed sowing, Water spray and Fertigation," 2022 International Conference on Computational Intelligence and Sustainable Engineering Solutions (CISES), 2022, pp. 148-153, doi: 10.1109/CISES54857.2022.9844341.
- [19] Rakhra, Manik, Partho Deb, Omdev Dahiya, Sahil Sonu Chandel, Brinderjit Bhutta, Sumit Badotra, Sunny Kumar, Aman Shaukat, and Dalwinder Singh. "An Analytical Study of the Types of Implements used by Farmers in Mechanized Agriculture." In 2022 International Mobile and Embedded Technology Conference (MECON), pp. 683-687. IEEE, 2022.
- [20] Shruti, R. Soumya, M. Rakhra, N. S. Chintagunti, B. Singh and D. Singh, "Modern Data Mining Approach to Handle Multivariate Data and to Implement Best Saving Services for Potential Investor," 2022 International Mobile and Embedded Technology Conference (MECON), 2022, pp. 611-616, doi: 10.1109/MECON53876.2022.9752101.
- [21] M. N. Gowda, D. Singh and M. Rakhra, "Machine Learning – Based Diagnosis of Covid-19 using Clinical Data," 2022 3rd International Conference on Intelligent Engineering and Management (ICIEM), 2022, pp. 910-916, doi: 10.1109/ICIEM54221.2022.9853083.
- [22] Chhabra, Abhishek, Tenzin Woeden, Dalwinder Singh, Manik Rakhra, Omdev Dahiya, and Aditya Gupta. "Image Steganalysis with Image decoder using LSB and MSB Technique." In 2022 3rd International Conference on Intelligent Engineering and Management (ICIEM), pp. 900-905. IEEE, 2022.